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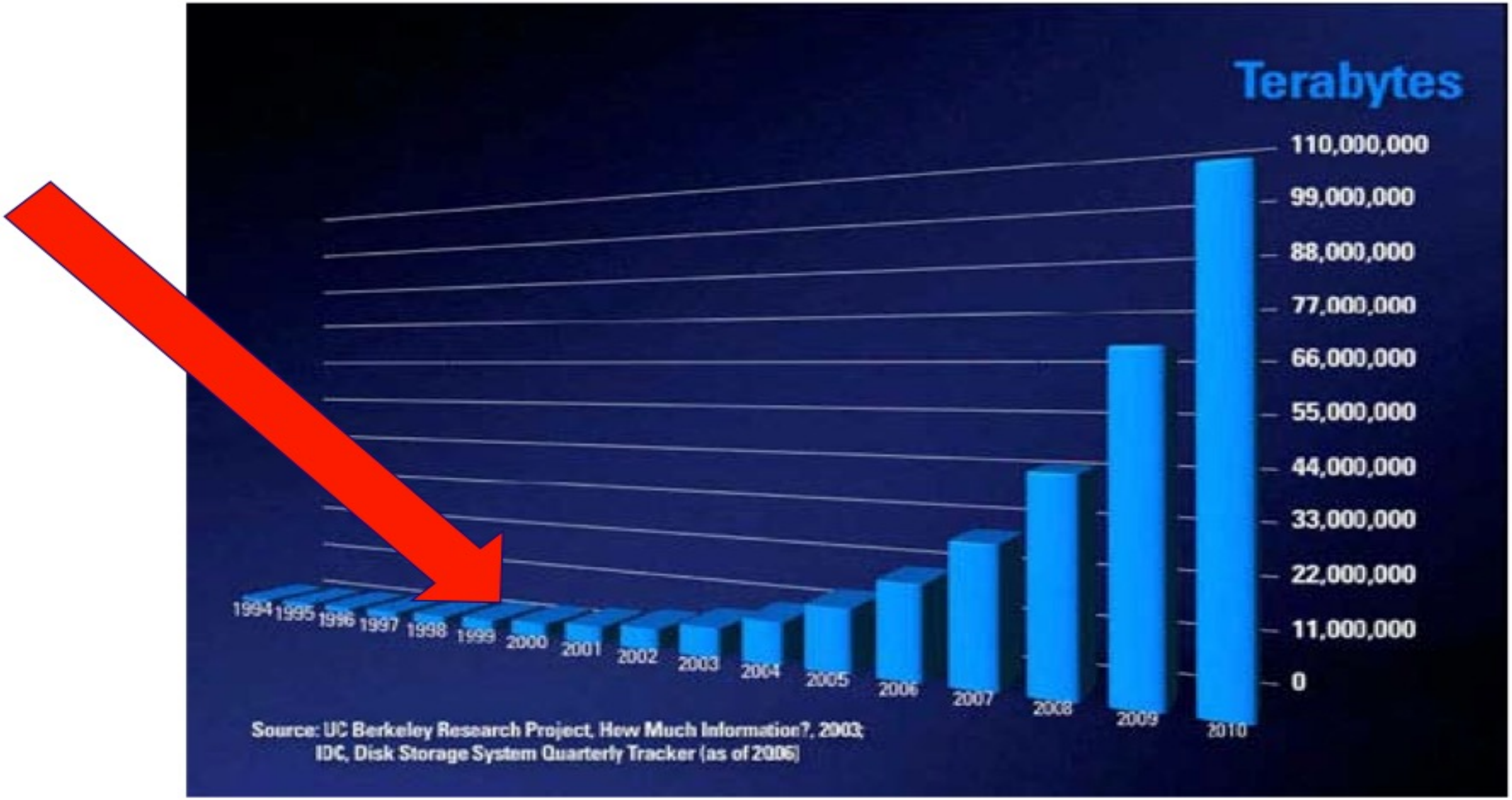
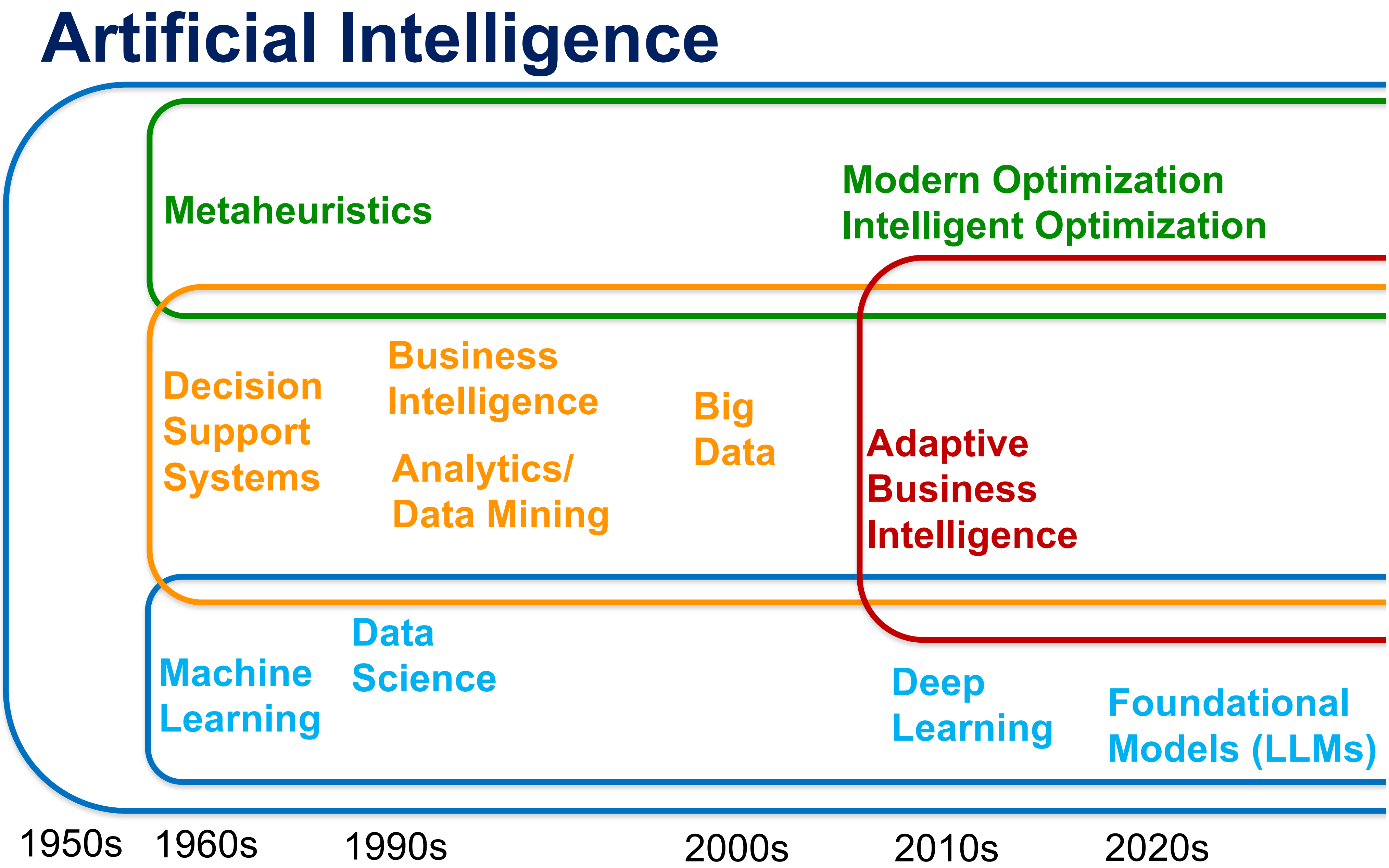
Q1-2

The Age of Big Data (and Data Science, Analytics, Data Mining, Business Intelligence, Decision Support Systems, Artificial Intelligence)!



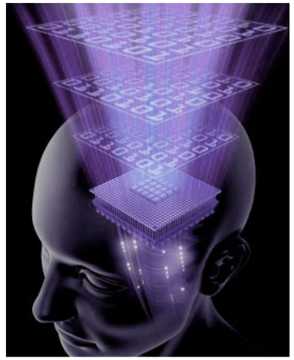


Artificial Intelligence
Metaheuristics
Modern Optimization
Intelligent Optimization
Decision Support Systems
Business Intelligence
Analytics/ Data Mining
Big Data
Adaptive Business Intelligence
Machine Learning
Data Science
Deep Learning
Foundational Models (LLMs)
1950s
1960s
1990s
2000s
2010s
2020s

Modern Intelligent Decision Support Systems

Figure 7: Arnotts' genealogy about the DSS, adapted from Arnott and Pervan (2014)

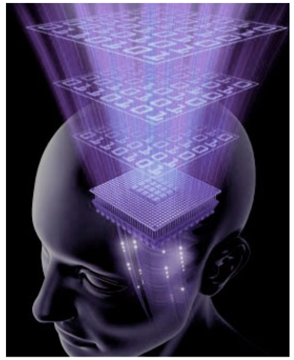
> Business Intelligence (BI)



> Motivation

- Nowadays, it is easy to **collect, store and process** data;
- The amount of stored data doubles every 9 months;
- Data can hold important information to aid decision making;
- Human experts are limited!

> Business Intelligence (BI)



> Definition

- **Business Intelligence (BI)** is an umbrella term that includes methodologies, architectures, tools, applications and technologies to enhance managerial decision making.
- The goal of **BI** is to: access data from multiple sources, transform these data into information, knowledge and **actions**.
- **Current trends**: real-time access to data (get the right data at the right time and place), use of Web data (BI 2.0), sensor data (BI 3.0), ….

> Adaptive Business Intelligence (ABI)

> What do organizations need?

- Answer to 2 basic questions:
  - What is more likely to happen in the **future**?

    Prediction/Forecasting (Data Mining)
  - What is the best possible **decision given** what is known now?

    Optimization (Modern Optimization Methods)

> Adaptive Business Intelligence (ABI)

> Examples

- To get to X, I have to predict traffic patterns and then optimize the best traffic path;

- To invest, I need to know what is going to happen in the market and then identify the best investment strategy.

> Adaptive Business Intelligence (ABI)

> Definition

- BI System that combines adaptive optimization and adaptive forecasting in order to answer the 2 basic questions;
- Thus, **ABI software** should include:
- (**Adaptive) Forecasting System** and;
- (**Adaptive) Optimization System**.

> Adaptive Business Intelligence (ABI)

> Before ABI

- BI solutions that extract knowledge from data (know the past);
- Exemple: analysing the data of an organization, leading to tables, graphs and statistics like:
  - 57% of the consumers are within 40 to 50 years old;
  - Produt X has better sales in Lisbon than in Oporto.

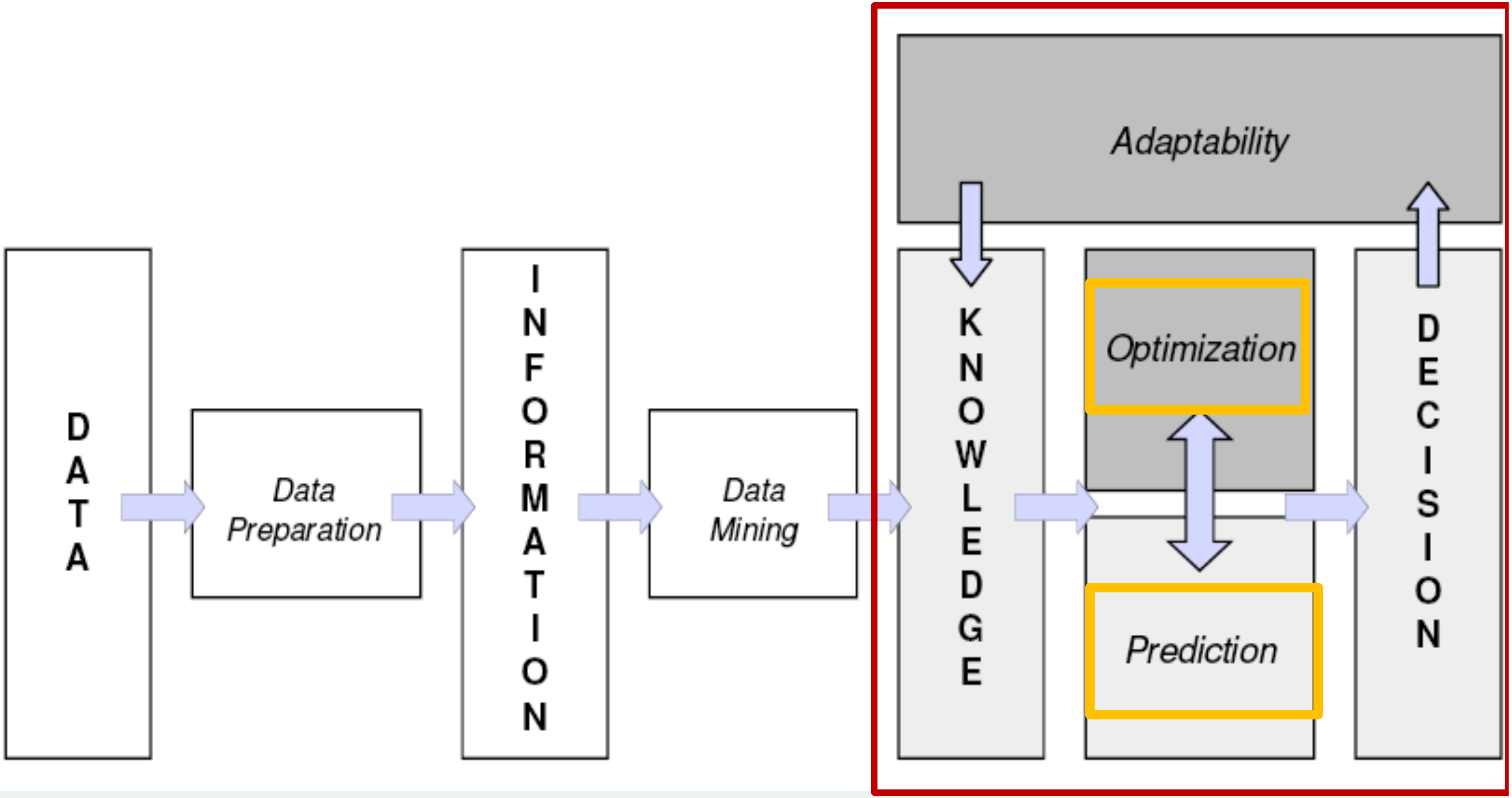
So what???

> Adaptive Business Intelligence (ABI)

> However:

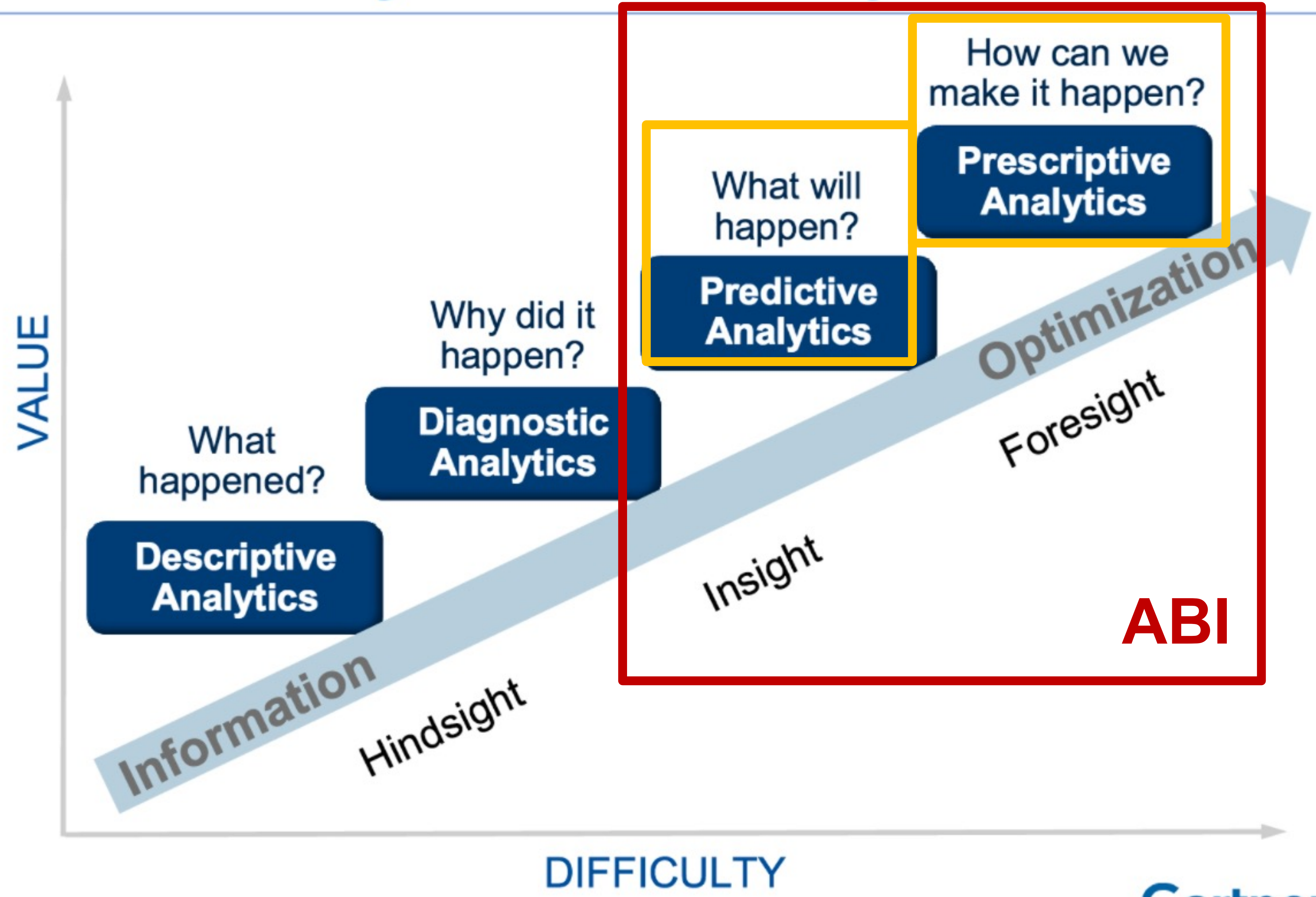
- Knowledge is not useful if it is not used to aid/support decision making! Or, in other words, if it does not result in an **ACTION**!

Adaptive Business Intelligence (ABI)



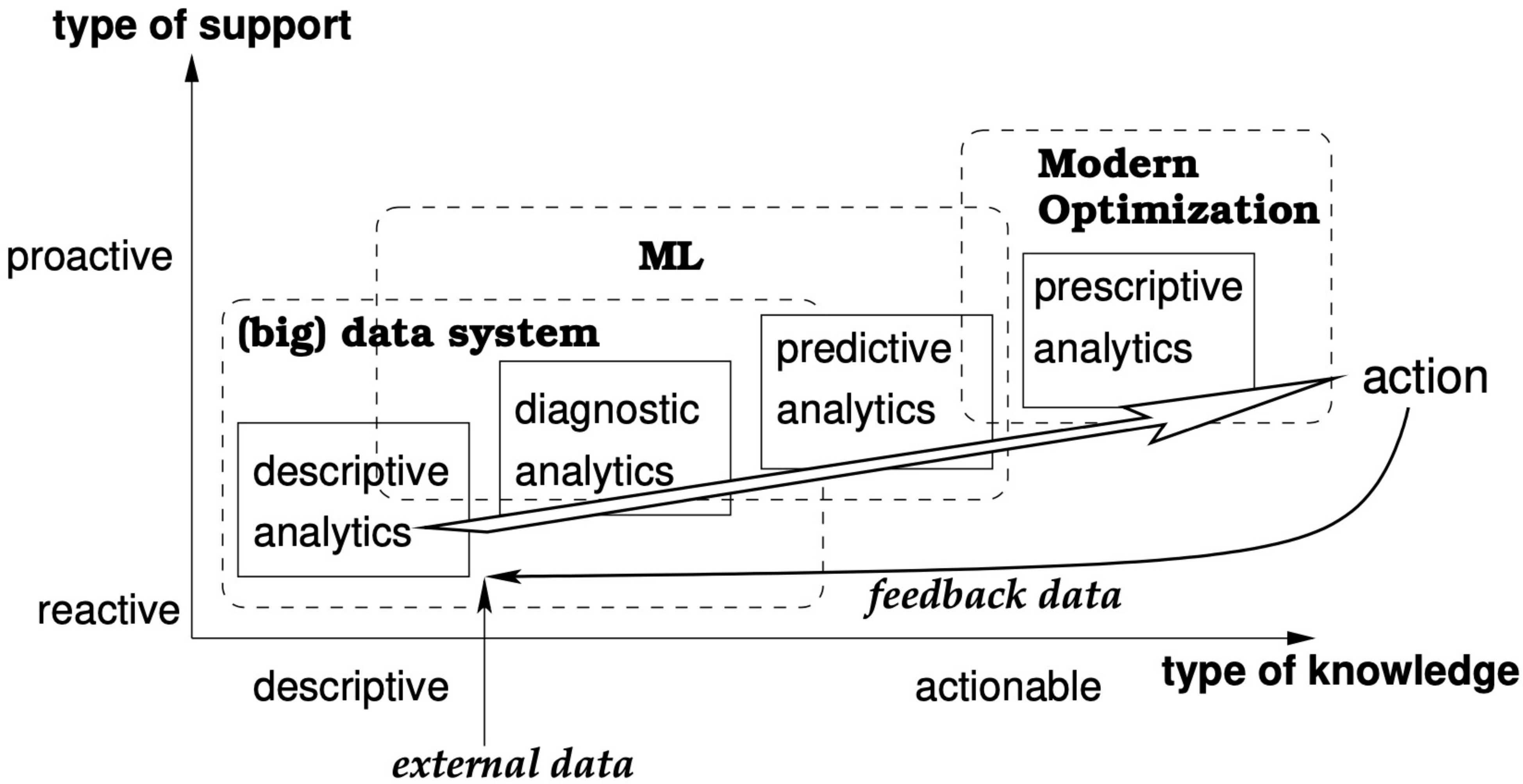
ABI adds **intelligent** features to Business Intelligence (BI);
ABI has been applied in several applications: marketing campaigns, investment strategies, credit card fraud detection.

Gartner Analytic Ascendancy Model



Gartner

TADA (Technology for Adaptive Data Analytics)



> Complex Business Problems

- **Very difficult to solve (basically, all easy problems have already been solved);**
- Includes all kinds of real-world problems:
    - Scheduling problems;
    - Marketing Optimization;
    - Stock trading;
    - Forecasting financial indexes;
    - Optimal water control;

> Complex Business Problems

▪Common characteristics:

▪The number of possible **solutions is so large** that it precludes a complete search for the best answer

▪The problem exists in a **time-changing environment**.

▪The problem is **heavily constrained**.

▪There are **many (possibly conflicting) objectives**.

▪**Incomplete data** to base decisions on

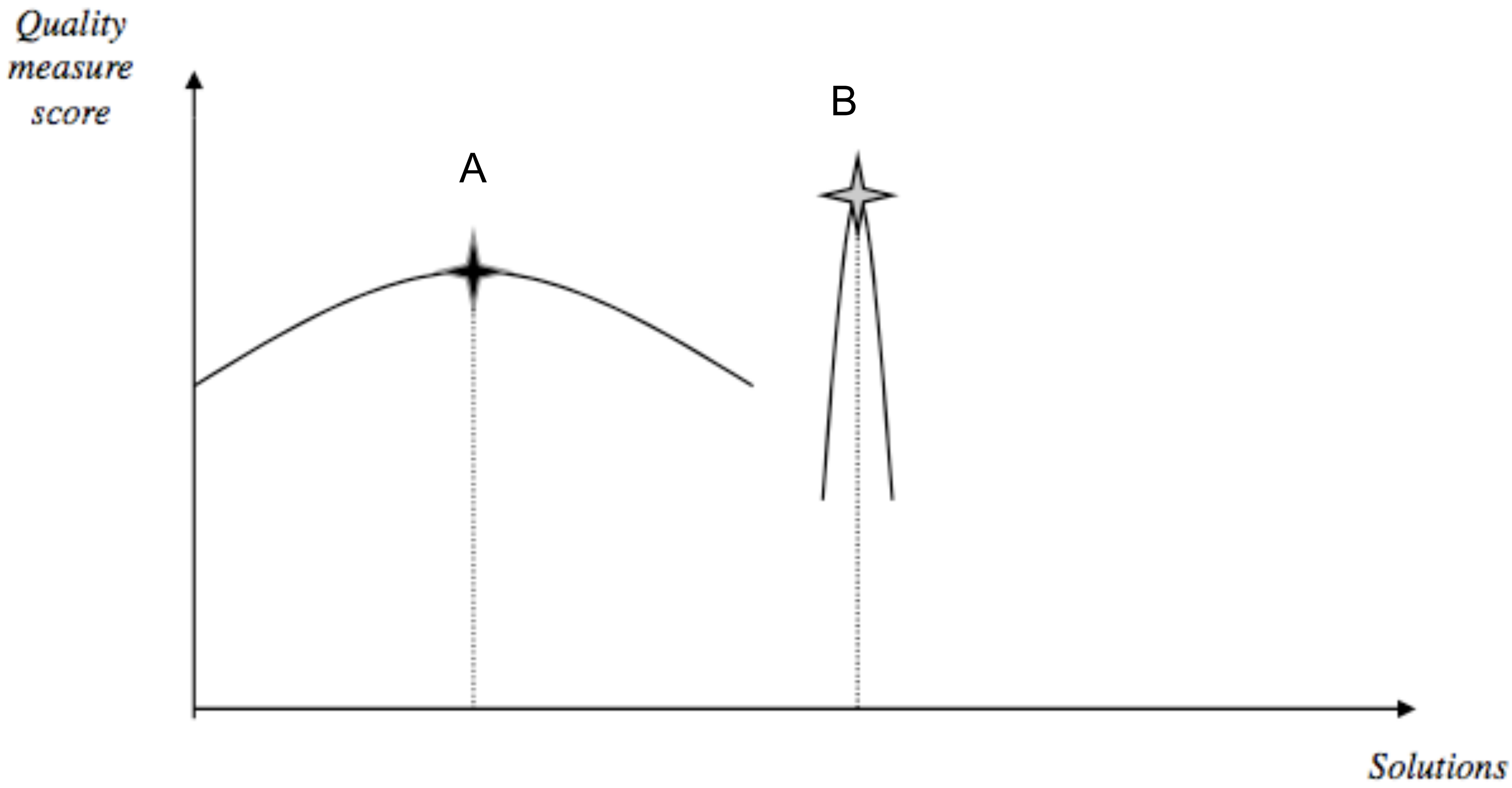
▪"Noisy" data, which typically refers to 'human errors' when entering information.

> Complex Business Problems

> Number of possible solutions:

- With 1 production line and 50 orders, the number of possible solutions is 50! =+- 10^{64}
- Brute force (blind search) requires too much computational effort;
- **Solution: better optimization strategy**

- **Dynamic environment (time):**
- What is the best solution?



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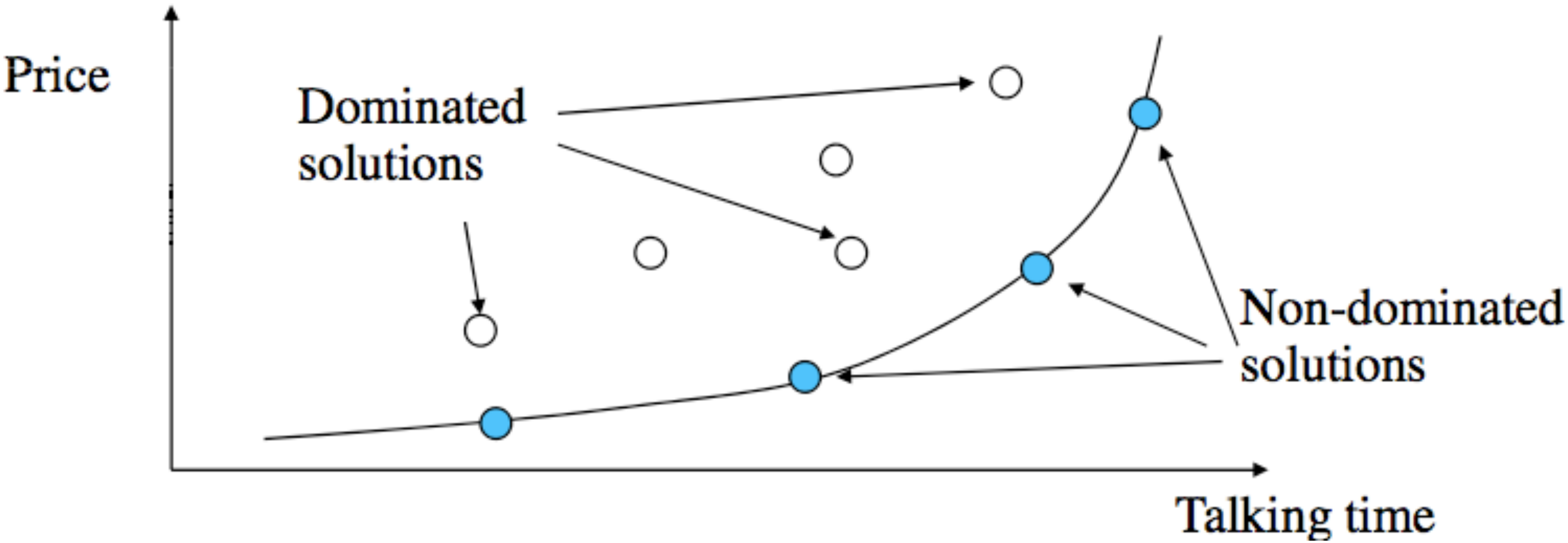
Q3

> Complex Business Problems

> Multi-objective problems:

- Maximize the talking time while minimizing the price of a cellular phone call;



> Complex Business Problems

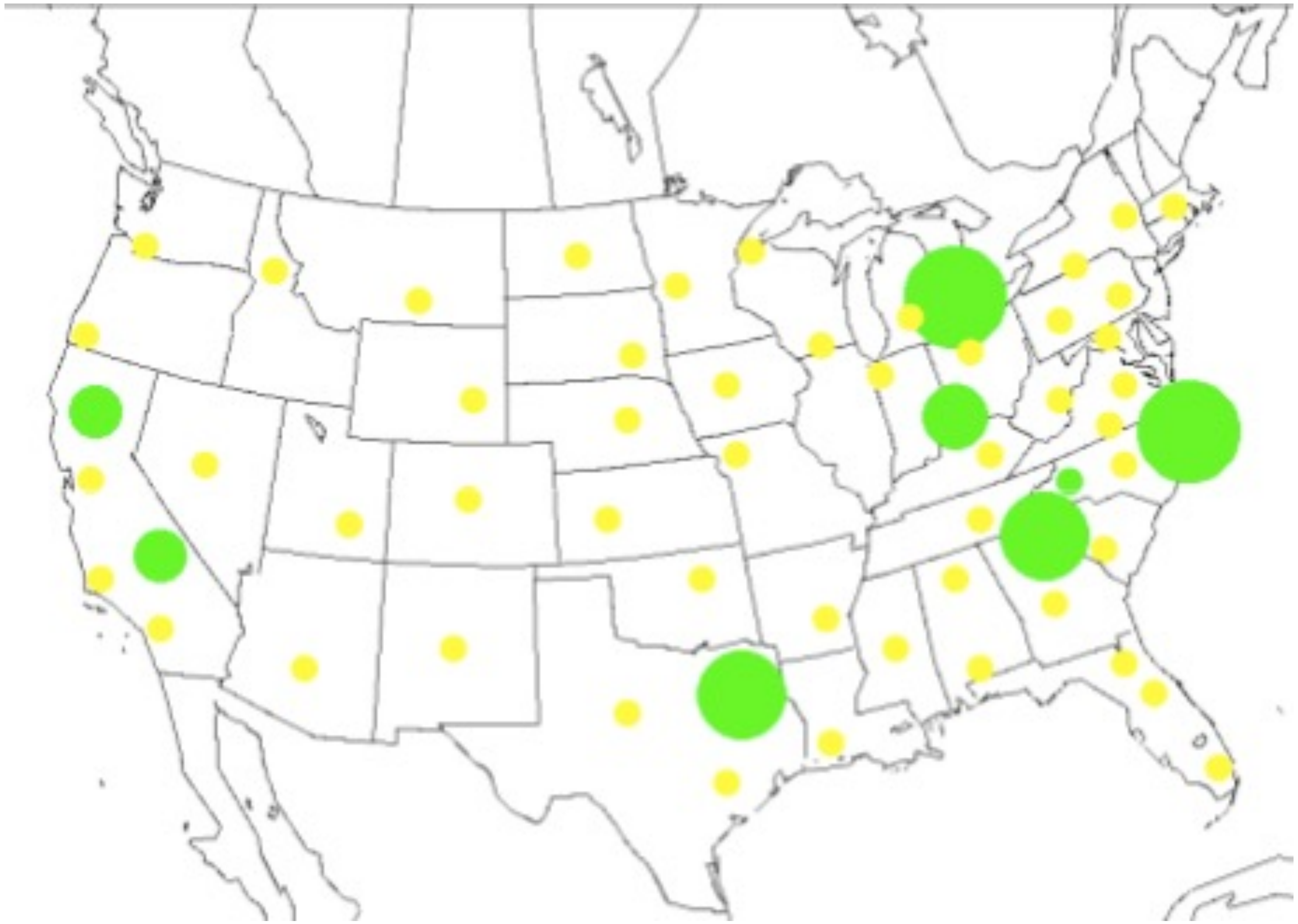
> ABI approach:

- Conventional Operational Research approach use a simplified model and search for the optimum solution;
- **ABI:** use a **model closer to the real-world** and search for an approximate solution (may not be the optimum);

> Car Distribution Example

> Characteristics:

- Thousands of leased cars are received by a leasing company;
- **Green point**: where cars are returned;
- **Yellow point:** where cars need to be sold at (auction point);



> Car Distribution Example

> AIM:

- Every day, the leasing company has to decide **where to send** each car received;
- Example: 3000 cars per day, 50 auction sites.
- **Simple Solution**: send one car at the time to the auction site where we expect the highest sales value.

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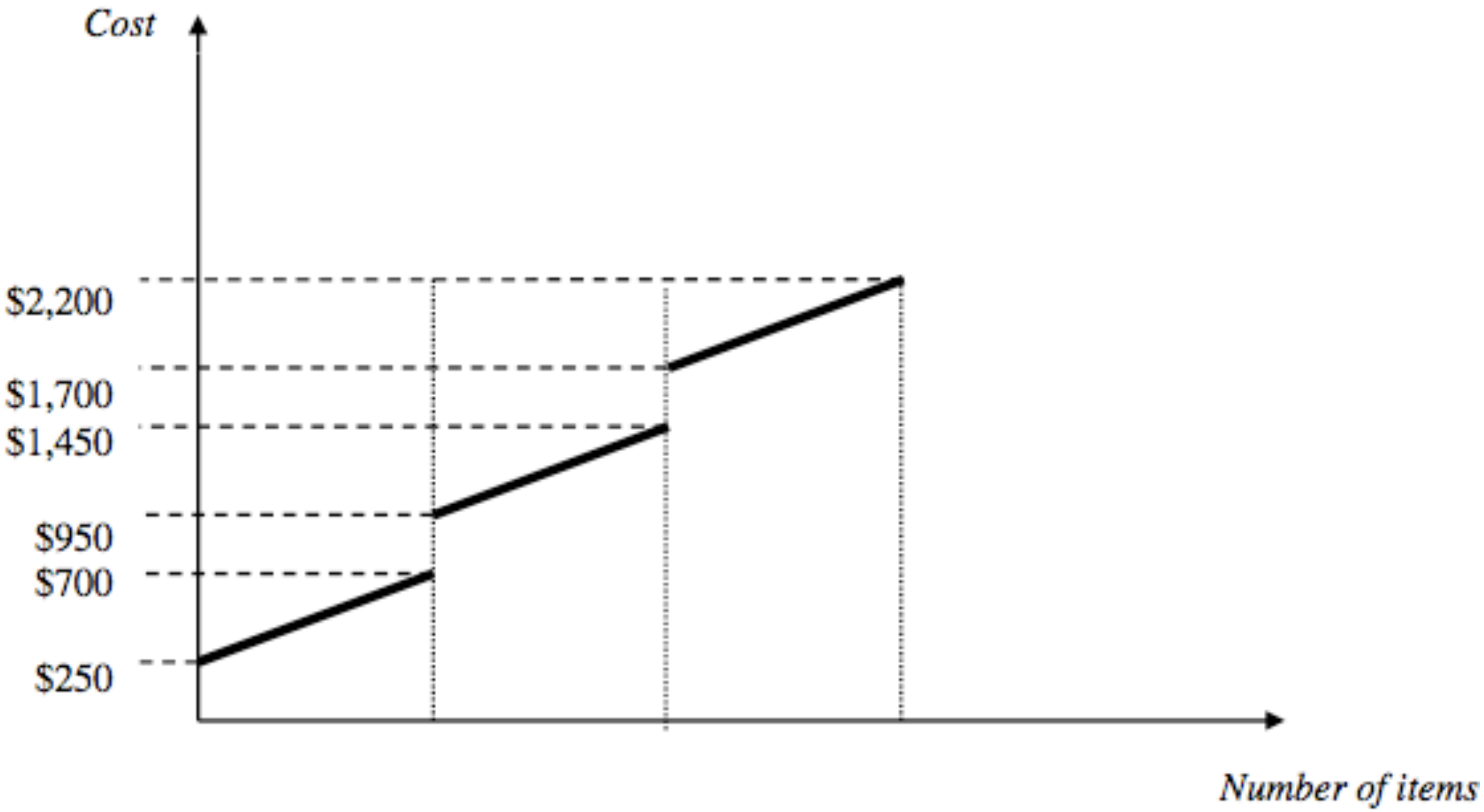
Q4

> Car Distribution Example

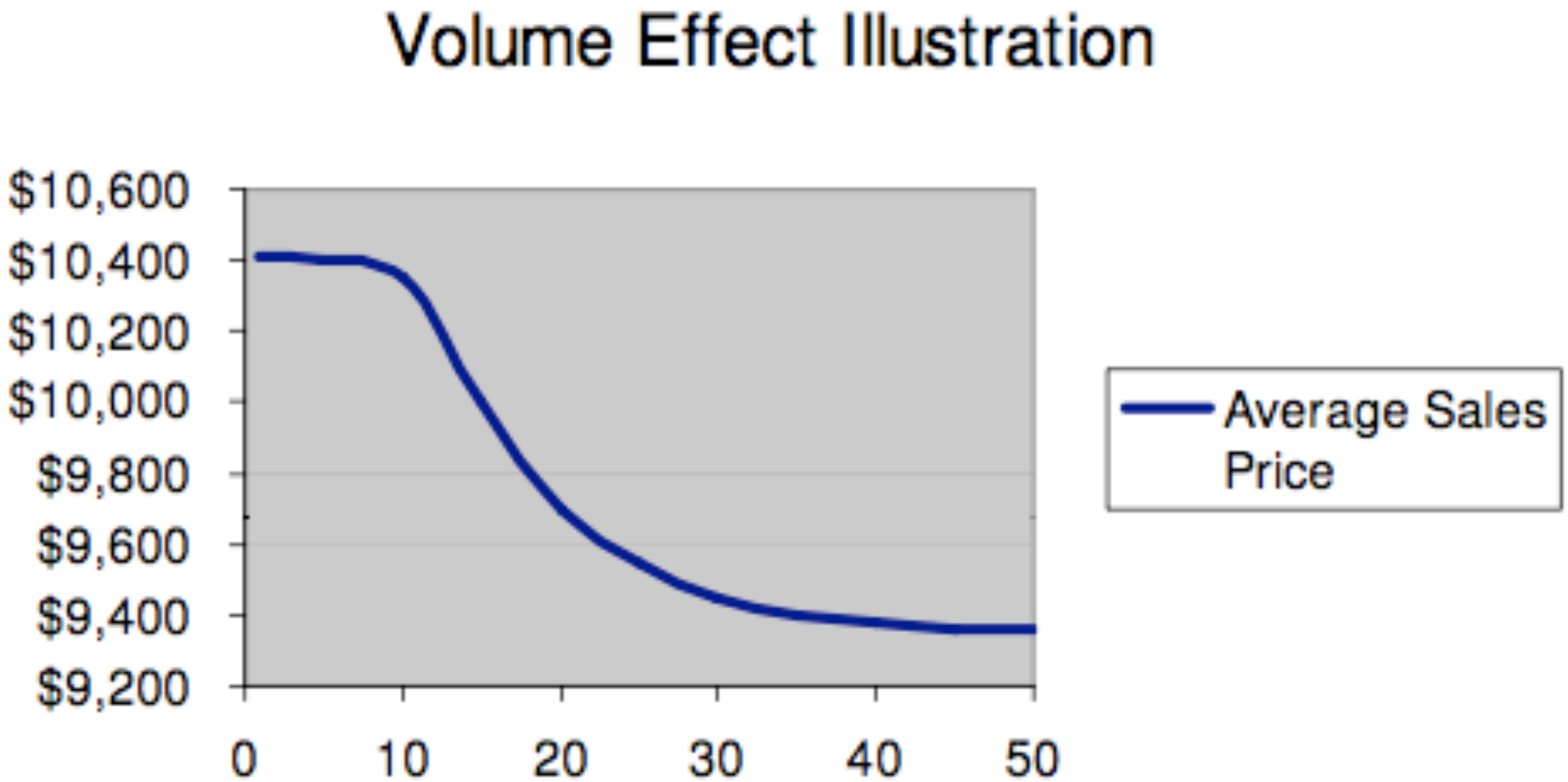
> Simple solution would lead to a disaster!

- On average, **it takes 27 days** for a car to reach the auction site;
- **Transportation cost** is not linear (batches into trucks):



> Car Distribution Example

- **Volume effect** (1 white ford mustang vs 100!)
- Taking one car at the time is not a good option!



> Car Distribution Example

- A car looses about $10 USD in value each day while it remains not sold.
- The **price depreciation** is based on the fact that if the car was sold quickly, the money could be invested and earn interests.
- Hence, we would like to **sell the cars as quickly as possible**, and the transportation time (and distance) should be minimized.
- Example: We might have 1000 cars at a particular auction site, but only 250 of our cars are sold (on average) at each sale.

> Car Distribution Example

- **Changes in the environment:** transportation costs, auction sale prices, new auction sites, summer vs winter (e.g. convertible), ...
- **Even with perfect information, the problem is complex due to:** volume effect, transport costs, price depreciation and all dynamic changes (time, risk factors, special events like auction in Las Vegas using only white cars, ...).

> Car Distribution Example

> ABI Impact:

- **Current solution:** human team specialized in shipping cars

to auction sites (very complex task);

- A small error of $150 per car implies hundreds of thousands of

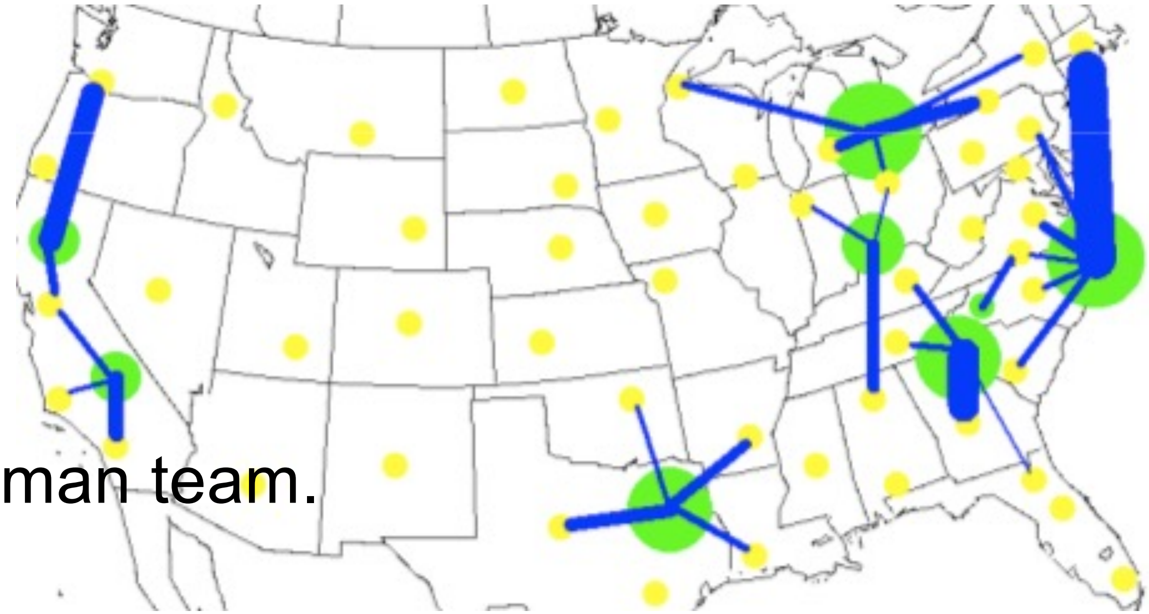
$ per day!

- Michalewicz ABI system:

estimate of an improvement

of several million $ per year

When compared with the human team.



> Car Distribution Example

> Prediction Module:

- **Based on historical sale records:**

  f(base_price,model,mileage,season,other)

Example:

if [Make = Honda] **and** [Model = Accord] **and** [Color = white] **and** [40,000 < Mileage < 50,000] **and** [Year = 2000] **and** [Damage Level = $0], **then** Sale Price = $11,384.

> Car Distribution Example

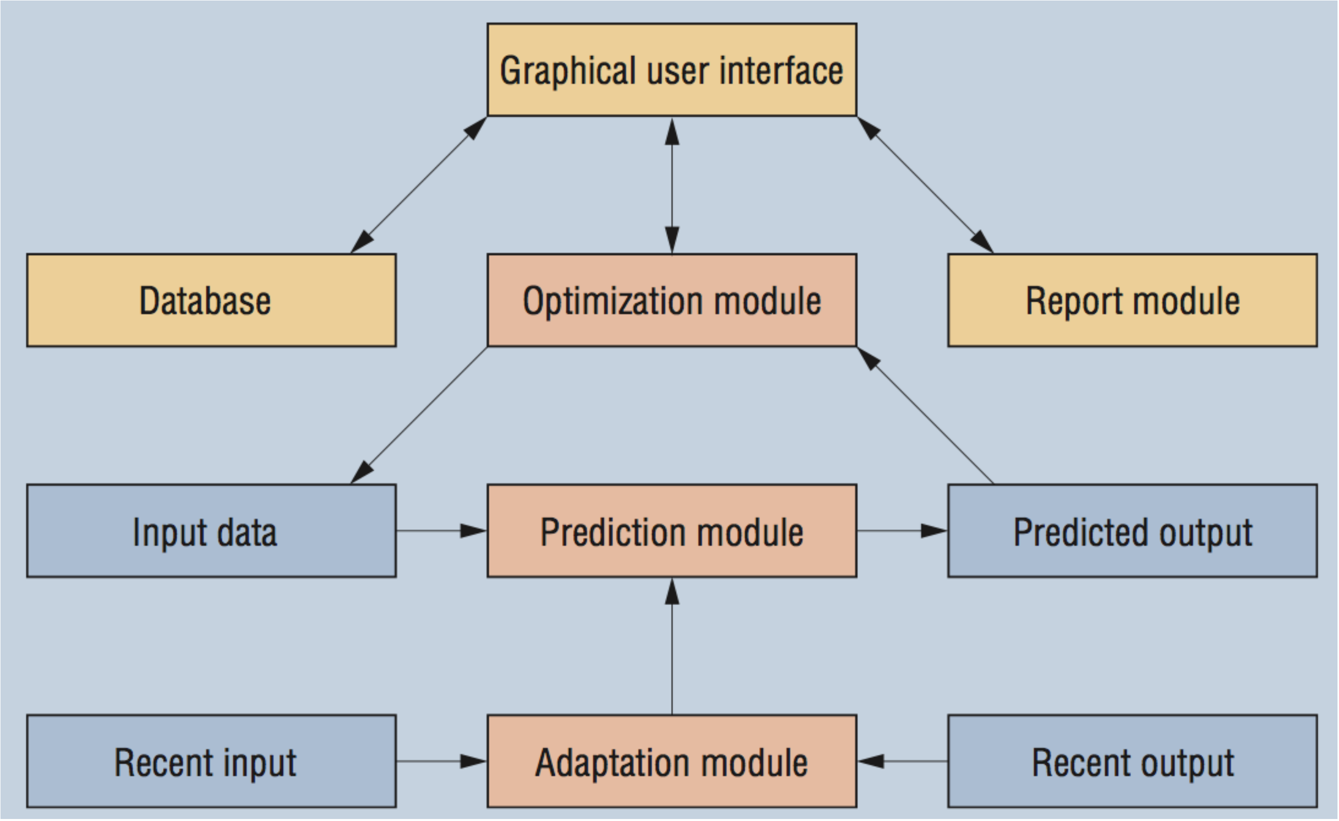
> Optimization Module:

- Based on an **evolutionary algorithm**;
- Each solution defines for where to send (auction site) the N daily cars (N=2000);
- Each solution is **evaluated** for its “quality”:
  - It calculates the expected sales (adjusted to volume and price depreciation effects);
  - Subtracts costs related to a car distribution (transportation costs, auction fees,…);

> Car Distribution Example

> Adaptability:

- Use of a **rolling window** to update (adapt) the prediction module (every month);



> Other Applications

- **ABI Michalewicz book:** marketing campaigns, investment strategies, credit card fraud, …
- Several of my research papers (to be shown to the students)

> Perusall exercise for this week

Create a Perusall account at (use your institutional email and then only your ID at any of the name field, such as: a32456, repeat the ID if needed):

https://perusall.com/

Course code to enroll:

-HFGCZ

Direct enroll access:

https://app.perusall.com/join/-HFGCZ

> Perusall exercise for this week

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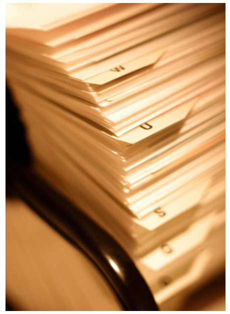
> Bibliography



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